

TANVI PATIL

Mumbai, Maharashtra

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Education

Sardar Patel Institute of Technology

Bachelor of Science in Computer Science (CGPA: 8.45)

Nov. 2022 – May 2026

Mumbai, Maharashtra

MatoShree Prabodhini High School

Higher Secondary Education - HSC Board (91.7%)

Sep. 2020 – May 2022

Mumbai, Maharashtra

A. K. Joshi School

Secondary Education - SSC Board (97.6%)

Sep. 2010 – May 2020

Mumbai, Maharashtra

Technical Skills

Programming Languages: Python, JavaScript, HTML/CSS, SQL

Frameworks and Technologies: Flask, Node.js, Express.js, React.js, React Native, MySQL, MongoDB

Tools and Libraries: Microsoft Excel, VBA, Pandas, NumPy, Scikit-learn, Selenium

Platforms: Solved 300+ LeetCode problems 🐙, GitHub 🐙

Projects

Business Analytics with Python – KPI Modeling and Retail Insights | Python, Pandas, Scikit-learn 🐙 July 2023

- Processed **500,000+** retail transaction records using **machine learning algorithms** to identify performance drivers across categories and regions.
- Executed **customer segmentation** using **K-Means clustering** creating **3+** clusters and developed **random forest model** achieving **85% accuracy** for churn prediction.
- Designed **time series forecasting models** for sales prediction and conducted **cross-selling analysis** via **ABI matrix** and **correlation heatmaps**.

NLP-based Smart Travel Assistant – Travello | React Native, Flask, Node.js, LangChain, NLP 🗒️🐙 December 2024

- Engineered intelligent travel assistant using **LangChain** and **NLP** for **intent recognition**, generating personalized routes from natural language with **95% accuracy** using **OpenStreetMap** and **GTFS data**.
- Developed **one-click ticketing system** serving **10+** transport modes with **real-time GTFS** and **OpenStreetMap APIs**, optimized via **Gunicorn** and **RabbitMQ**.

FundVisor: Secure Crowdfunding Platform | React Native, Node.js, Python, MongoDB 🐙

February 2025

- Architected secure crowdfunding platform with **escrow-based fund release** and **milestone-based payments** using **RazorPay API**, processing **\$50K+** transactions.
- Created **founder credibility verification** using **ML reinforcement algorithms** achieving **78% risk assessment accuracy** to reduce investor risk and enhance trust.

SignLingual – Real-time Sign Language Translator | Python, CNN, OpenCV 🐙

August 2024

- Constructed **real-time sign language translator** using **Convolutional Neural Networks** achieving **92% accuracy** for hand gesture classification from webcam input across **26 alphabets**.
- Trained **custom ML models** on **10,000+** gesture datasets and integrated **OpenCV** for **real-time text conversion** processing **30 FPS** to enhance accessibility.

Achievements

- 1st Place:** VESIT Hackathon, IIIT BHU Hackathon, Shark-A-Thon (RGIT)
- 2nd Place:** Innovation Oasis (IIIT Nagpur), Ideathon (VESIM), Case Competition (SCOE)
- 3rd/4th Place:** Unicus (VJTI), Brandstruck (IIT Bombay), PAN-IIT PIWOT Hackathon (Jio World Centre)

Leadership Experience / Extracurricular

Technical Lead (Virtual Stock Market - VSM)

December 2023 – May 2024

SPIT

- Spearheaded technical operations and app development, ensuring smooth platform performance and seamless event execution.

Member, Rotaract Club

July 2022 – June 2023

SPIT

- Coordinated **15+** donation drives and **8 campus sustainability initiatives**, engaging **500+** students in community service programs promoting social responsibility.